UNIVERSITY OF THE PEOPLE

CS 2203-01 - AY2026-T4 Programming

Assignment Activity Unit 6

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Introduction

Database Management Systems (DBMS) are essential for organizing, storing, and retrieving data efficiently. Structured Query Language (SQL) allows users to create database structures, manipulate records, and manage relationships between entities. In this assignment, the existing library management database from Unit 4 was extended by adding an Authors table and populating the database with sample records. SQL queries were also implemented to retrieve and manipulate data within the system.

Source Code: https://github.com/mubashir1837/MAY-7/blob/main/library_database.sql

Part A: Schema Extension

The existing database schema was extended by creating a new table called Authors with the attributes AuthorID, Name, Nationality, and BirthYear.

SQL Query for Authors Table

```sql
CREATE TABLE Authors (
    AuthorID INT PRIMARY KEY,
    Name VARCHAR(100),
    Nationality VARCHAR(50),
    BirthYear INT
);
```

Existing Tables

Books Table

```sql
CREATE TABLE Books (
    ISBN VARCHAR(20) PRIMARY KEY,
    Title VARCHAR(100),
    Author VARCHAR(100),
    Genre VARCHAR(50),
    Quantity INT
);
```

Members Table

```sql
CREATE TABLE Members (
    MemberID INT PRIMARY KEY,
    Name VARCHAR(100),
    Email VARCHAR(100),
    Phone VARCHAR(20)
);
```

Loans Table

```sql
CREATE TABLE Loans (
    LoanID INT PRIMARY KEY,
    MemberID INT,
    ISBN VARCHAR(20),
    LoanDate DATE,
    ReturnDate DATE
);
```

Part B: Populating Tables with Sample Data

Insert Records into Authors Table (5 Records)

```sql
INSERT INTO Authors VALUES
(1, 'J.K. Rowling', 'British', 1965),
(2, 'George Orwell', 'British', 1903),
(3, 'Paulo Coelho', 'Brazilian', 1947),
(4, 'Dan Brown', 'American', 1964),
(5, 'Harper Lee', 'American', 1926);
```

Insert Records into Books Table (10 Records)

```sql
INSERT INTO Books VALUES
('ISBN001', 'Harry Potter', 'J.K. Rowling', 'Fantasy', 10),
('ISBN002', '1984', 'George Orwell', 'Dystopian', 8),
('ISBN003', 'Animal Farm', 'George Orwell', 'Political Fiction', 6),
('ISBN004', 'The Alchemist', 'Paulo Coelho', 'Adventure', 12),
('ISBN005', 'Inferno', 'Dan Brown', 'Thriller', 7),
('ISBN006', 'Digital Fortress', 'Dan Brown', 'Tech Thriller', 5),
('ISBN007', 'To Kill a Mockingbird', 'Harper Lee', 'Classic', 9),
('ISBN008', 'Brida', 'Paulo Coelho', 'Fiction', 4),
('ISBN009', 'Angels and Demons', 'Dan Brown', 'Mystery', 11),
('ISBN010', 'Fantastic Beasts', 'J.K. Rowling', 'Fantasy', 13);
```

Insert Records into Members Table (20 Records)

```sql
INSERT INTO Members VALUES
(1,'Ali Khan','ali@gmail.com','03001111111'),
(2,'Ahmed Noor','ahmed@gmail.com','03002222222'),
(3,'Sara Ali','sara@gmail.com','03003333333'),
(4,'Fatima Khan','fatima@gmail.com','03004444444'),
(5,'Usman Tariq','usman@gmail.com','03005555555'),
(6,'Hina Malik','hina@gmail.com','03006666666'),
(7,'Bilal Ahmed','bilal@gmail.com','03007777777'),
(8,'Ayesha Noor','ayesha@gmail.com','03008888888'),
(9,'Zain Ali','zain@gmail.com','03009999999'),
(10,'Hamza Khan','hamza@gmail.com','03110000000'),
(11,'Mariam Asif','mariam@gmail.com','03111111111'),
(12,'Tariq Mahmood','tariq@gmail.com','03112222222'),
(13,'Noor Fatima','noor@gmail.com','03113333333'),
(14,'Asad Ali','asad@gmail.com','03114444444'),
(15,'Iqra Khan','iqra@gmail.com','03115555555'),
(16,'Saad Ahmed','saad@gmail.com','03116666666'),
(17,'Laiba Noor','laiba@gmail.com','03117777777'),
(18,'Hassan Ali','hassan@gmail.com','03118888888'),
(19,'Anaya Khan','anaya@gmail.com','03119999999'),
(20,'Umar Farooq','umar@gmail.com','03220000000');
```

Part C: SQL Queries

1. Retrieve All Books Written by a Specific Author

SELECT * FROM Books

WHERE Author = 'Dan Brown';

Expected Result

ISBN	Title	Author	Genre	Quantity
ISBN005	Inferno	Dan Brown	Thriller	7
ISBN006	Digital Fortress	Dan Brown	Tech Thriller	5
ISBN009	Angels and Demons	Dan Brown	Mystery	11

2. Drop the Authors Table

DROP TABLE Authors;

3. Identify Names of Members Who Borrowed a Specific Book

```
74   INSERT INTO Loans VALUES
75   (1, 1, 'ISBN001', '2026-05-01', '2026-05-10'),
76   (2, 2, 'ISBN005', '2026-05-02', '2026-05-11'),
77   (3, 3, 'ISBN005', '2026-05-03', '2026-05-12');
78
```

```sql
SELECT * FROM Books
WHERE Author = 'Dan Brown';

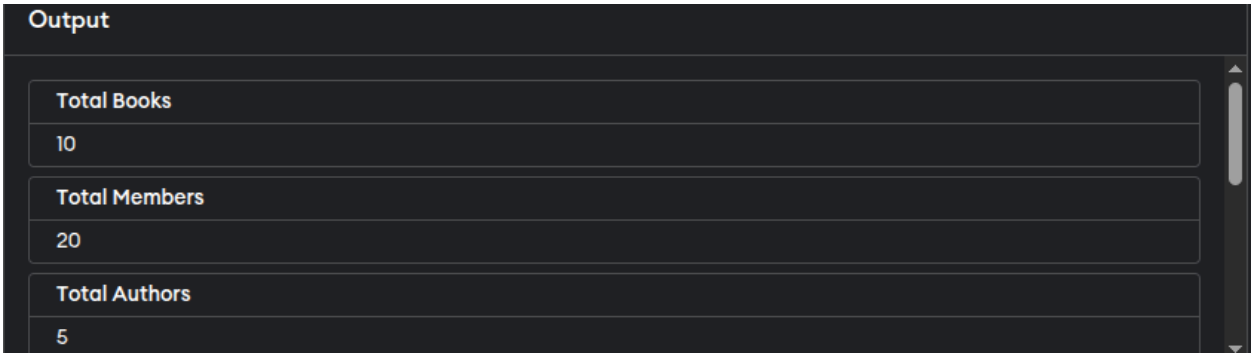

DROP TABLE Authors;


SELECT DISTINCT Members.Name, Books.Title
FROM Members
INNER JOIN Loans ON Members.MemberID = Loans.MemberID
INNER JOIN Books ON Loans.ISBN = Books.ISBN
WHERE Loans.ISBN = 'ISBN005';


ALTER TABLE Members
ADD MembershipType VARCHAR(20);

```

Result



4. Alter Members Table to Add MembershipType

```sql
ALTER TABLE Members
ADD MembershipType VARCHAR(20);

```

Conclusion

This assignment demonstrated the practical implementation of SQL Data Definition Language (DDL) and Data Manipulation Language (DML) commands within a library database system. The database schema was successfully extended using the CREATE TABLE command, and sample data was inserted into multiple tables using INSERT statements. Furthermore, queries such as SELECT, DROP, and ALTER TABLE were used to retrieve information, remove database objects,

and modify existing table structures. These operations are essential for maintaining accurate and efficient database systems in real-world applications such as libraries, educational systems, and business environments.

References

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